

STA 2002

Lecture 14

Midterm Review

Midterm Topics

- Point estimation: maximum likelihood estimator, unbiased estimator, method of moments
- Confidence intervals for means, variances and proportions
- Hypothesis tests for means, variances and proportions
- Power and sample size
- ~~Order statistics~~

Point estimation

You should be able to:

- Know CLT and MGF from previous courses
- Know the common rules of derivative/integration
- understand the difference between population parameters like μ,σ and sample statistics like $\bar{x}, s$
- derive the Maximum Likelihood Estimator for a parameter
- derive the Method of Moments Estimator for a parameter
- determine if an estimator is unbiased or not

Estimation

Let $X_1, \ldots, X_n$ be iid with pdf

$$f(x|\theta) = \theta x^{\theta-1}, \quad 0 \le x \le 1, \quad 0 < \theta < \infty.$$

(a) Find the MLE of θ

$$\prod_i \theta x_i^{\theta-1} = \theta^n \left(\prod_i x_i\right)^{\theta-1} = L(\theta|\mathbf{x})$$

$$\frac{d}{d\theta} \log L = \frac{d}{d\theta}\left[n\log\theta + (\theta-1)\log\prod_i x_i\right] = \frac{n}{\theta} + \sum_i \log x_i.$$

$$\hat{\theta} = (-\tfrac{1}{n}\textstyle\sum_i \log x_i)^{-1}$$

Estimation

Find the method of moments estimator of θ.

$$\mathrm{E}(X) = \int_0^1 x\theta x^{\theta-1} dx = \frac{\theta}{\theta+1} \int_0^1 (\theta+1) x^{\theta} dx = \frac{\theta}{\theta+1}$$

$$\frac{1}{n} \sum_i X_i = \frac{\theta}{\theta+1}$$

$$\tilde{\theta} = \frac{\sum_i X_i}{n - \sum_i X_i}$$

Estimation

Let $f(x;\theta) = (1/\theta)x^{(1-\theta)/\theta}, \quad 0 < x < 1, \quad 0 < \theta < \infty.$

(a) Show that the maximum likelihood estimator of θ is $\widehat{\theta} = -(1/n)\sum_{i=1}^{n} \ln X_i$.

$$
\begin{aligned}
L(\theta) &= \left(\frac{1}{\theta^n}\right)\left(\prod_{i=1}^{n} x_i\right)^{1/\theta-1}, \qquad 0<\theta<\infty \\
\ln L(\theta) &= -n\ln\theta + \left(\frac{1}{\theta}-1\right)\ln\prod_{i=1}^{n} x_i \\
\frac{d\ln L(\theta)}{d\theta} &= \frac{-n}{\theta} - \frac{1}{\theta^2}\ln\prod_{i=1}^{n} x_i = 0 \\
\widehat{\theta} &= -\frac{1}{n}\ln\prod_{i=1}^{n} x_i \\
&= -\frac{1}{n}\sum_{i=1}^{n}\ln x_i.
\end{aligned}
$$

Estimation

6.1-20. Let the p.d.f. of X be defined by

$$f(x) = \begin{cases} \left(\frac{4}{\theta^2}\right)x, & 0 < x \leq \frac{\theta}{2}, \\ -\left(\frac{4}{\theta^2}\right)x + \frac{4}{\theta}, & \frac{\theta}{2} < x \leq \theta, \\ 0, & \text{elsewhere,} \end{cases}$$

where $\theta \in \Omega = \{\theta : 0 < \theta \leq 2\}$.

(a) Sketch the graph of this p.d.f. when $\theta = 1/2$, $\theta = 1$, and $\theta = 2$.

(b) Find an estimator of θ by the method of moments.

(c) For the following observations of X, give a point estimate of θ:

0.3206	0.2408	0.2577	0.3557	0.4188
0.5601	0.0240	0.5422	0.4532	0.5592

(a)

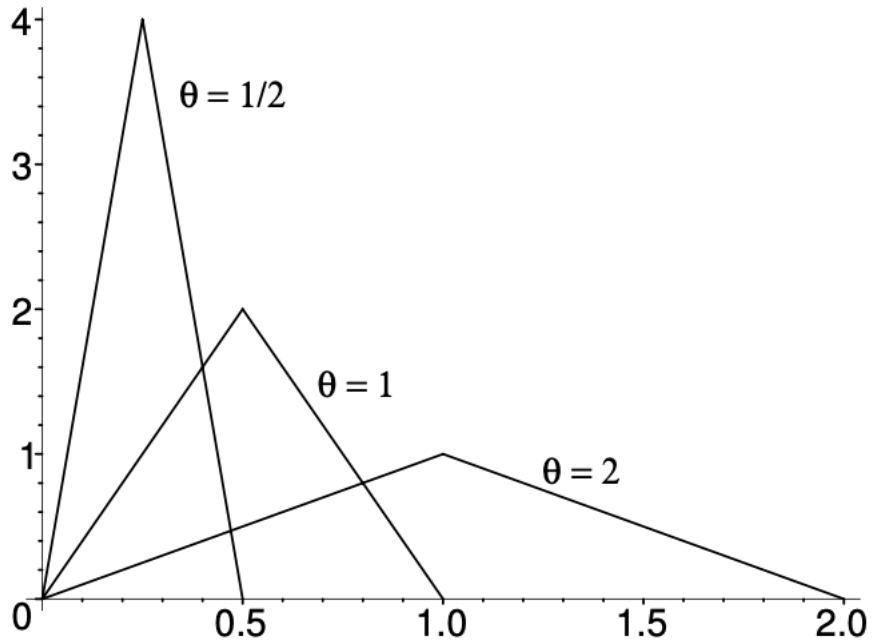


Figure 6.1–20: The p.d.f.s of X for three values of θ

(b) $E(X) = \theta/2$. Thus the method of moments estimator of θ is $\widetilde{\theta} = 2\overline{X}$.

(c) Since $\overline{x} = 0.37323$, a point estimate of θ is $2(0.37323) = 0.74646$.

Confidence intervals

You should:

- know which confidence interval to use in a given setting
- know which assumptions are needed for the confidence interval to be correct

You learned how to find confidence intervals for

- A single population:
  - mean μ, proportion p, variance σ^2
- Two populations:
  - difference in means $\mu_1 - \mu_2$
    - If samples from the two populations are dependent (paired t-interval)
    - If samples from the two populations are independent (pooled t-interval if equal variances, Welch's t-interval if different variances)
  - difference in proportions $p_1 - p_2$
  - difference in variances σ_1^2/σ_2^2

Which assumptions are needed – example: Interval for μ

Always assume: $X_1, .., X_n$ are **i.i.d.** random variables from a distribution with mean μ, variance σ^2

a) $\bar{x} \pm z_{\alpha/2}\, \sigma/\sqrt{n}$

- Assumes: (1) σ is known, (2) distribution of $X_1, .., X_n$ is normal or n is large enough for the interval to be approximately correct by CLT

b) $\bar{x} \pm t_{\frac{\alpha}{2}}(n-1)\, s/\sqrt{n}$

- Assumes: distribution of $X_1, .., X_n$ is normal. If not, and n is large, then we can heuristically argue that the interval is approximately the same as the interval in (b), and thus an approximate C.I. by the CLT:
  - $t_{\frac{\alpha}{2}}(n-1) \approx z_{\frac{\alpha}{2}}$ if n is large
  - We also used $t_{n-1,\frac{\alpha}{2}}$ for $t_{\frac{\alpha}{2}}(n-1)$ in the notes.

c) $\bar{x} \pm z_{\alpha/2}\, s/\sqrt{n}$

- This is again an approximate C.I. if n is large, see argument in (b)

Hypothesis testing

Important concepts:

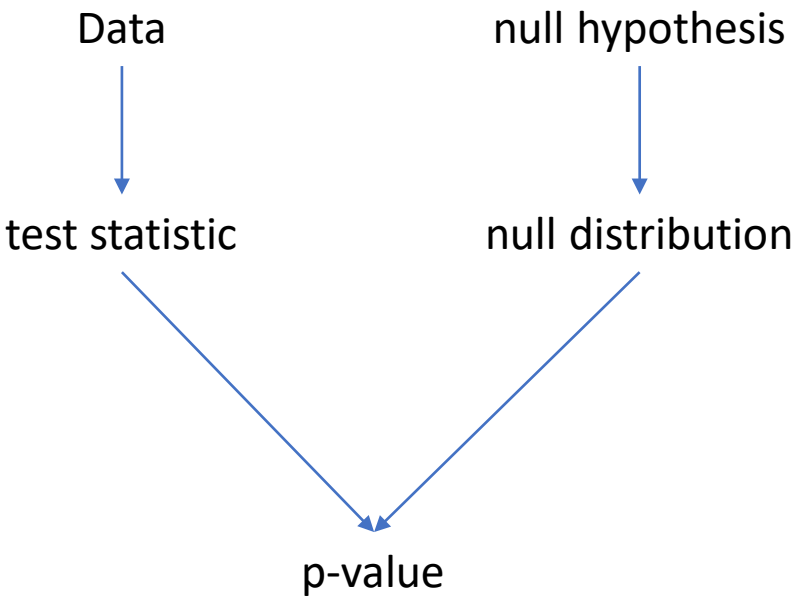
- Null hypothesis, alternative hypothesis
- Type I error (reject H_0 when H_0 is true)
- type II error (do not reject H_0 when H_1 is true)
- Test statistic (statistic for which we know the distribution under H_0)
- Critical region (if test statistic is in this region, we reject H_0)
- p-value (*probability under H_0 of a test outcome that is as or more favoring H_1 than the observed test statistic*)

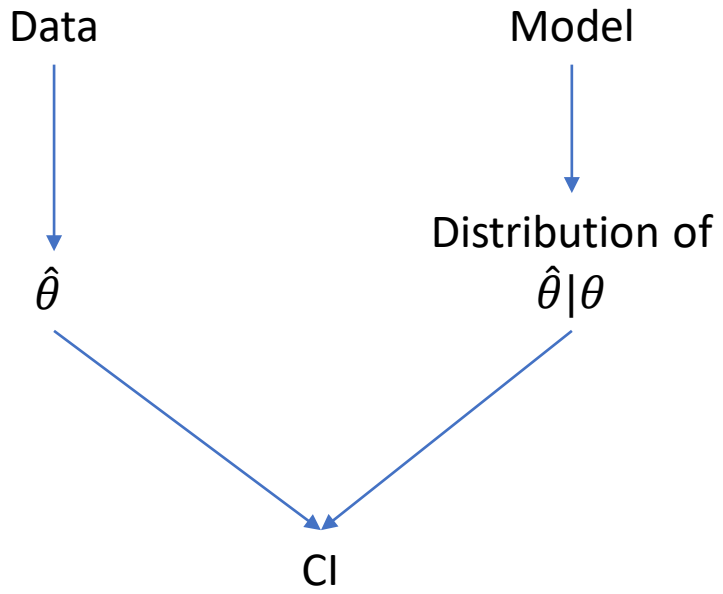
Hypothesis testing

You learned how to do hypothesis tests for

- A single population:
  - mean μ, proportion p, variance σ^2
- Two populations:
  - difference in means $\mu_1 - \mu_2$
    - If samples from the two populations are dependent (paired t-test, did not discuss in details)
    - If samples from the two populations are independent (pooled t-test if equal variances, Welch's t-test if different variances)
  - difference in proportions $p_1 - p_2$
  - difference in variances σ_1^2/σ_2^2 (F-test)

CI and tests


Data
null hypothesis
test statistic
null distribution
p-value


Data
Model
$\hat{\theta}$
Distribution of
$\hat{\theta}|\theta$
CI

8.2 In a given city it is assumed that the number of automobile accidents in a given year follows a Poisson distribution. In past years the average number of accidents per year was 15, and this year it was 10. Is it justified to claim that the accident rate has dropped?

$$P(X \leq 10|\,\lambda = 15) = \sum_{i=0}^{10} \frac{e^{-15}\,15^i}{i!} = e^{-15}\left[1+15+\frac{15^2}{2!}+\cdots+\frac{15^{10}}{10!}\right] \approx .11846.$$

Normal approximation X~N(15,15)

$$z = \frac{10.5 - 15}{\sqrt{15}} = -1.16$$

One-sided p=0.122

CI and tests

A 95% CI for mean is constructed using 64 samples. The CI is:

[6, 14]

What is the sample mean and sample variance?

$$\left[\bar{x} - t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right), \bar{x} + t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right)\right]$$

$\bar{x} = 10$

$$\frac{2s}{\sqrt{64}} = 4$$

s=16

A major goal in early-stage clinical trials is to assess the rate of adverse effects. After the company has collected data from the first 100 subjects who received the drug, it conducts an interim analysis to assess the adverse event rate. Among the first 100 subjects, 10 reported some adverse effects after taking the drug, such as dizziness during the day, feeling tired all the time etc.

For the questions below, please show your deductions and reasons in detail.

(a). At the alpha level of 0.05, is the adverse effect rate significantly different than the average rate of 15%? State the hypotheses form clearly, and show your reasons.

$H_0 : p = 0.15$

$H_1 : p \neq 0.15$

$\frac{0.1-0.15}{\sqrt{0.15*0.85/100}} = -1.40$

Under level of significance $\alpha = 0.05$, not significant because $|-1.40| < 1.96$.

(b). Suppose the incidence rate is less than 15%. With 100 subjects, what is the maximum number of adverse events, such that the test in item (a) will be significant?

$$\frac{\hat{p} - 0.15}{\sqrt{0.15 * 0.85/100}} \leq -1.96$$

Thus, $\hat{p} \leq 0.15 - 1.96 * \sqrt{0.15 * 0.85/100} = 0.08001$

8 will make it significant.

(c). The company plans to collect more subjects in the treatment group, and assume the incidence rate is still 10%. What is the minimum number of subjects (in total) to make the test conducted in item (a) significant?

$\frac{0.1-0.15}{\sqrt{\frac{0.15*0.85}{n}}} = -1.96$

n=195.9

So, 196.